
Canvas LMS

Canvas provides a comprehensive notification system that overlaps with several features planned in SACP. Users can receive notifications through multiple delivery channels including email, in-app activity stream, and mobile push notifications via the Canvas mobile app.

Overlapping Features:

1. Email Notifications

   Canvas supports both individual per-notification emails and daily digest emails, allowing users to choose how frequently they receive updates — similar to SACP's email & notifications service.

2. Targeted Messaging ("Message Students Who")

   Instructors can send targeted messages to specific student groups based on academic data — for example, students who have not submitted, submitted on time, or have ungraded assignments. This overlaps with SACP's targeted/personalized communication based on academic data.

3. Per-User Notification Preferences

   Each user can configure their notification frequency per channel

(immediately, daily, weekly, or never) and per notification type (due dates, grading, course content, file uploads, discussions, announcements). This aligns with SACP's role-based communication and user-level control.

4. Push Notifications

Canvas delivers push notifications to mobile devices via its mobile app, overlapping with SACP's planned real-time notification delivery.

5. Account-Level Global Notifications

Administrators can broadcast global notifications to specific roles (e.g., StudentEnrollment), with configurable start and end dates. This partially overlaps with SACP's campaign scheduling and role-based communication features.

6. API-Driven Notification Control

Canvas exposes a REST API for notification preferences, allowing programmatic control over channels and frequencies — relevant to SACP's backend-driven notification architecture.

Comparison Table:

Feature	SACP	Canvas

Email Notifications | YES | YES

In-App / Stream Notifications | YES | YES

Push Notifications (Mobile) | YES | YES

Role-Based Targeting | YES | YES

Academic-Data-Based Targeting | YES | LTD

Per-User Notification Prefs | YES | YES

Scheduled/Recurring Notifications| YES | NO

Campaign Scheduling / Bulk Msgs | YES | NO

Rule-Based Workflow Automation | YES | NO

Dynamic Message Templates | YES | NO

Delivery Tracking & Logs | YES | NO

Analytics & Engagement Reporting | YES | NO

AI/ML Email Classification | YES | NO

Real-Time 1-to-1 Chat | YES | NO

LTD = Limited / partial support

Missing Features:

- Rule-Based Workflow Automation: Canvas has no engine to automatically trigger notifications based on academic events such as attendance drops, missed deadlines, or exam schedules. SACP automates these using configurable rule-based workflows.

- Campaign Scheduling & Bulk Communication: Canvas does not provide

a campaign management interface. SACP allows administrators and faculty to create, schedule, and send bulk communication campaigns to targeted audiences.

- Dynamic Message Templates: Canvas has no template management system. SACP supports dynamic templates with personalized variables drawn from academic records.

- Delivery Tracking & Communication Logs: Canvas does not offer delivery tracking or a centralized communication audit trail. SACP logs all sent communications with delivery status.

- Analytics & Engagement Reporting: Canvas provides no dashboard for tracking notification open rates, delivery rates, or engagement metrics. SACP includes a real-time analytics dashboard.

- AI/ML Email Classification & Prioritization: Canvas has no intelligent filtering or prioritization of messages. SACP applies machine learning to classify and prioritize academic emails, reducing noise for students and faculty.

References:

- Instructure (Canvas Official) - https://www.instructure.com/en-gb/canvas-how-it-compares

- Canvas LMS REST API — Notification Preferences - https://canvas.instructure.com/doc/api/notification_preferences.html

- Canvas LMS REST API — Account Notifications - https://canvas.instructure.com/doc/api/account_notifications.html

- Canvas Notification Preferences Guide (Tufts) - https://tuftsedtech.screenstepslive.com/s/18992/m/73355/l/810714-how-do-i-set-my-canvas-notification-preferences

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Moodle LMS

Moodle is an open-source LMS with a flexible notification and messaging system. Its notification architecture shares several concepts with SACP's planned features.

Overlapping Features:

1. Multi-Channel Notification Delivery

   Moodle supports web (in-platform), email, and mobile push notifications. Push notifications are delivered via the AirNotifier server to iOS (APNs) and Android (FCM) devices — overlapping with SACP's multi-channel notification delivery plan.

2. Per-User Messaging Preferences

Users can configure which notification types they receive and through which channel (web, email, or mobile). Administrators set institution-wide defaults. This aligns with SACP's role-based and user-level notification control.

3. Bulk Messaging

Moodle provides a "course:bulkmessaging" capability that allows authorized users to send a message to many people simultaneously, overlapping with SACP's bulk email/message campaign feature.

4. Real-Time Messaging

Moodle includes a personal messaging system for direct one-to-one and group conversations between users, which overlaps with SACP's real-time one-to-one communication module.

5. Event-Based Notifications (via Plugin)

The Event Reminders plugin enables calendar-event-based notifications, partially overlapping with SACP's academic-event-triggered automated notifications.

6. Custom Notification API

Moodle's Message API allows plugins to send custom push notifications with user opt-in/opt-out control — similar to SACP's programmatic notification service architecture.

Comparison Table:

Feature	SACP	Moodle
Email Notifications	YES	YES
In-App / Stream Notifications	YES	YES
Push Notifications (Mobile)	YES	YES
Real-Time 1-to-1 Chat	YES	YES
Role-Based Targeting	YES	YES
Bulk Messaging	YES	YES
Per-User Notification Prefs	YES	YES
Campaign Scheduling	YES	NO
Rule-Based Workflow Automation	YES	NO
Dynamic Message Templates	YES	NO
Delivery Tracking & Logs	YES	NO
Analytics & Engagement Reporting	YES	NO
AI/ML Email Classification	YES	NO
Scheduled/Recurring Notifications	YES	NO

Missing Features:

- Rule-Based Workflow Automation: Moodle has no native engine to trigger automated notifications based on custom academic rules (e.g., attendance below threshold, assignment overdue). SACP's automation engine handles this without manual intervention.

- Campaign Scheduling & Management: Moodle has no campaign interface. SACP enables scheduled bulk communication campaigns with audience targeting and delivery management.

- Dynamic Message Templates: Moodle has no template system for personalized messages. SACP uses dynamic templates populated with data from academic records.

- Delivery Tracking & Analytics: Moodle does not track message delivery rates, read status, or engagement. SACP provides a dedicated analytics dashboard for communication performance.

- AI/ML Email Classification: Moodle applies no intelligent prioritization to messages. SACP uses NLP and ML models to classify and filter academic emails.

- Calendar reminders require external plugin installation and are not available out of the box, unlike SACP's built-in event-triggered notification system.

References:

- Moodle Docs — Notifications - https://docs.moodle.org/405/en/Notifications

- Moodle Docs — Messaging Settings - https://docs.moodle.org/39/en/Messaging_settings

- Moodle Docs — Mobile App Notifications - https://docs.moodle.org/502/en/Moodle_app_notifications

- Moodle Developer Resources — Custom Push Notifications - https://moodledev.io/general/app/development/custom-push-notifications

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Blackboard Learn

Blackboard (now under Anthology) is one of the most established LMS platforms in higher education. Its notification system is mature and covers multiple delivery methods with strong institutional controls.

Overlapping Features:

1. Multi-Channel Notification Delivery
   Blackboard supports activity stream (in-platform), email (individual or daily digest), mobile push notifications via the Blackboard app, and legacy SMS. This overlaps with SACP's multi-channel notification delivery.

2. Event-Driven Notifications
   The Blackboard notification framework automatically fires notifications when system events occur — assignment created,

grade posted, survey submitted, test overdue — without requiring manual sends. This partially overlaps with SACP's academic-event-triggered notification automation.

3. Role-Based & Course-Level Targeting

Notifications can be configured per course, per organization, or globally. Institutions can override individual user settings.
This aligns with SACP's role-based communication and targeting.

4. Email Digest & Frequency Control

Users choose between individual emails per notification or a daily digest. This overlaps with SACP's email notification service and delivery control.

5. Push Notifications

The Blackboard mobile app delivers push notifications for course events, messages, and grade updates — overlapping with SACP's real-time notification delivery.

6. In-Course Messaging

Blackboard's Course Messages feature enables direct messaging between instructors and students within the course environment, overlapping with SACP's real-time one-to-one communication.

Comparison Table:

Feature	SACP	Blackboard
Email Notifications	YES	YES
In-App / Stream Notifications	YES	YES
Push Notifications (Mobile)	YES	YES
SMS Notifications	-	LEGACY
Real-Time 1-to-1 Chat	YES	YES
Role-Based Targeting	YES	YES
Per-User Notification Prefs	YES	YES
Event-Driven Notifications	YES	YES
Campaign Scheduling / Bulk Msgs	YES	NO
Rule-Based Workflow Automation	YES	NO
Dynamic Message Templates	YES	NO
Delivery Tracking & Logs	YES	NO
Analytics & Engagement Reporting	YES	NO
AI/ML Email Classification	YES	NO

Missing Features:

- Rule-Based Custom Workflow Automation: Although Blackboard fires notifications on system events, it does not support custom rule-based workflows (e.g., "if attendance < 75%, send warning to student and copy advisor"). SACP's automation engine handles such configurable academic rules.

- Campaign Management: Blackboard has no campaign creation, scheduling, or bulk communication management interface. SACP provides a full campaign module for faculty and administrators.

- Dynamic Message Templates: Blackboard has no reusable dynamic template system with academic data variables. SACP supports parameterized templates drawn from student/course records.

- Communication Analytics Dashboard: Blackboard provides no engagement metrics, delivery tracking, or open-rate reporting. SACP's analytics dashboard covers all of these.

- AI/ML Email Prioritization: Blackboard applies no intelligent classification to messages or notifications. SACP uses ML models to filter spam and prioritize important academic communications.

- SMS is being phased out (legacy only, pre-August 2023 institutions). SACP should consider adding SMS support via a modern gateway to cover this gap.

References:

- Blackboard Help — Notifications for Students - https://help.blackboard.com/Learn/Student/Original/Stay_in_the_Loop/Notifications

- Blackboard Help — About the Notifications System - https://help.blackboard.com/Learn/Administrator/Hosting/Tools_Management/Notifications/Understanding_the_Framework/About_the_Notifications_System

- Blackboard Help — Set Up Notifications (Ultra) - https://help.blackboard.com/Learn/Instructor/Ultra/Courses/Set_Up_Notifications/Notifications_Settings

- Blackboard Help — Course Messages - https://help.blackboard.com/Blackboard_App/Course_Messages

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Ellucian Banner (ERP)

Ellucian Banner is a widely used Student Information System (SIS) and ERP in higher education. Its communication capabilities are focused on institutional and administrative messaging rather than real-time collaboration.

Overlapping Features:

1. Email Notifications via Communication Management

   Banner's Communication Management module supports email template creation, scheduling, and delivery — overlapping with SACP's

email notification service and campaign scheduling.

2. Mobile Push Notifications

Ellucian GO and Ellucian Mobile deliver push notifications to iOS (APNs) and Android (FCM) devices. Institutions can configure push notifications for holds, restrictions, and custom messages from external apps via a RESTful Notifications API — overlapping with SACP's push notification and API-driven delivery plans.

3. Dynamic Message Templates

Banner Communication Management supports email, mobile notification, and letter templates with data fields populated from the SIS. Templates can be tested before publishing and scheduled for delivery — directly overlapping with SACP's dynamic template management feature.

4. Campaign / Bulk Message Scheduling

Administrators can create messages from published templates, schedule delivery within a valid date range, and target specific audiences based on SIS data — overlapping with SACP's campaign creation and scheduling module.

5. Role-Based & SIS-Data-Based Targeting

Banner uses SIS records (student status, holds, enrollment data) to target communications — closely aligned with SACP's academic-

data-based targeting feature.

6. In-App Notification Center

Banner 9's UI includes a Notification Center that displays real-time system alerts to logged-in users, overlapping with SACP's in-platform notification stream.

Comparison Table:

Feature	SACP	Banner
Email Notifications	YES	YES
Push Notifications (Mobile)	YES	YES
In-App Notifications	YES	YES
Dynamic Message Templates	YES	YES
Campaign Scheduling / Bulk Msgs	YES	YES
Academic-Data-Based Targeting	YES	YES
Role-Based Communication	YES	YES
Scheduled/Recurring Notifications	YES	YES
Real-Time 1-to-1 Chat	YES	NO
Rule-Based Workflow Automation	YES	NO
Delivery Tracking & Logs	YES	LTD
Analytics & Engagement Reporting	YES	NO
AI/ML Email Classification	YES	NO
Per-User Notification Prefs	YES	LTD

LTD = Limited / partial support

Missing Features:

- Real-Time One-to-One Communication: Banner has no real-time chat or messaging between users. SACP provides a live WebSocket-based messaging module for students, faculty, and administrators.

- Rule-Based Workflow Automation: Banner does not support custom event-triggered notification workflows (e.g., auto-notify when attendance drops). Communication campaigns must be manually initiated by administrators. SACP automates these workflows.

- Communication Analytics Dashboard: Banner does not provide delivery tracking, open rates, or engagement metrics for sent communications. SACP's analytics module covers all of these.

- AI/ML Email Classification: Banner applies no intelligent prioritization to communications. SACP's ML service classifies and filters academic emails to reduce noise.

- Self-Service for Faculty/Students: Banner's communication tools are primarily admin-facing. SACP gives faculty and students direct access to relevant communication features.

References:

- Ellucian Mobile — Set Up Notifications - https://mobile.elluciancloud.com/mobilehelp/c_set_up_notifications.html

- Banner Communication Management User Guide (Release 9.4) - https://banner9.unm.edu/communication_mgmt_9_6.pdf

- TrustRadius — Ellucian Banner Features - https://www.trustradius.com/products/ellucian-banner/features

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Oracle PeopleSoft Campus Solutions (ERP)

Oracle PeopleSoft Campus Solutions is an enterprise ERP used by

large universities worldwide. Its Notifications Framework is the

most technically robust among the systems reviewed here, and shares

the most architectural overlap with SACP.

Overlapping Features:

1. Multi-Channel Notifications Framework

   PeopleSoft supports Email, SMS, Push Notifications (iOS/Android),

   Portal Alerts, Worklist Items, and Announcements — all through

a single extensible framework. This overlaps with SACP's multi-channel notification service.

2. Event-Driven & Batch Notifications

The framework fires notifications in real-time when data changes or Application Engine processes complete, and also supports batch notification runs. This overlaps with SACP's academic-event-triggered automation workflows.

3. Scheduled & Recurring Notifications

Notification Publisher allows messages to be sent immediately, at a scheduled date/time, or as recurring events — directly overlapping with SACP's campaign scheduling feature.

4. Dynamic Templates with Ad-Hoc Text

Notifications use configurable templates with system-provided variables. Admins can also inject ad-hoc text at send time via a comment box. This overlaps with SACP's dynamic message templates.

5. Role & Audience Targeting

Notifications are routed by user role, user ID, or email address. Campus Mobile lets students manage their own preferences per notification type and channel — aligning with SACP's role-based targeting and per-user preferences.

6. SMS Support

PeopleSoft natively supports SMS as a notification channel, which is a channel SACP has not yet specified but should consider for high-priority alerts.

7. Reminder Notifications

The framework supports generating reminder notifications for existing pending notifications — overlapping with SACP's ability to send follow-up/reminder messages for deadlines and events.

Comparison Table:

Feature	SACP	PeopleSoft
Email Notifications	YES	YES
SMS Notifications	-	YES
Push Notifications (Mobile)	YES	YES
In-App / Portal Alerts	YES	YES
Worklist / Actionable Alerts	-	YES
Event-Driven Notifications	YES	YES
Batch Notifications	YES	YES
Scheduled/Recurring Notifications	YES	YES
Dynamic Message Templates	YES	YES

Role-Based Targeting | YES | YES

Reminder Notifications | YES | YES

Per-User Notification Prefs | YES | YES

Campaign Management Interface | YES | PARTIAL

Real-Time 1-to-1 Chat | YES | NO

Delivery Tracking & Logs | YES | NO

Analytics & Engagement Reporting | YES | NO

AI/ML Email Classification | YES | NO

Missing Features:

- AI/ML Email Classification & Prioritization: PeopleSoft has no machine learning layer for intelligent filtering or prioritization of messages. SACP's Python-based ML service addresses this gap, helping users focus on the most relevant communications.

- Real-Time One-to-One Chat: PeopleSoft has no live chat or direct messaging module. SACP provides WebSocket-based real-time messaging between students, faculty, and admins.

- Communication Analytics Dashboard: PeopleSoft has no dedicated reporting on notification delivery rates, read status, or engagement. SACP's analytics module provides real-time visibility into communication performance.

- End-User Friendly Campaign Interface: While PeopleSoft's
  Notification Publisher allows some scheduling, it requires
  developer/admin involvement to configure consumers and templates.
  SACP provides a self-service campaign management interface for
  faculty and administrators without technical knowledge.

- Centralized Communication Audit Logs: PeopleSoft does not provide
  a central log of all communications sent across the institution.
  SACP maintains comprehensive communication logs for accountability
  and review.

References:

- Oracle PeopleSoft — Understanding the Notifications Framework - https://docs.oracle.com/cd/E56917_01/cs9pbr4/eng/cs/lscc/concept_UnderstandingTheNotificationsFramework.html

- Oracle PeopleSoft — Getting Started With Campus Mobile - https://docs.oracle.com/cd/E56917_01/cs9pbr4/eng/cs/lsss/concept_GettingStartedWithCampusMobile-2d7ffe.html

- Oracle PeopleSoft — Understanding Notification Publisher - https://docs.oracle.com/en/applications/peoplesoft/peoplesoft-common/enterprise-components/understanding-notification-publisher.html

- Oracle PeopleSoft — Setting Up Online Notifications - https://docs.oracle.com/cd/E56917_01/cs9pbr4/eng/cs/lssr/task_SettingUpOnlineNotifications-3c8000.html

- Oracle PeopleSoft — Events and Notifications Framework (PDF) - https://docs.oracle.com/cd/F48196_01/psft/pdf/cs92eews-b102021.pdf